Software Testing and Quality Evaluation

Emergency Resource Coordination Platform

Group submission | SE3090 Assignment 2 | Revised 6 October 2026

This report evaluates the existing ASP.NET Core, PostgreSQL, React, Flutter and agentic subsystem developed for the SEF assignment. The verified aggregate is 270 passing automated cases, a passing local load scenario and a passing migration CLI check. Two security defects and the migration defect were corrected and retested. All current suites and the load scenario were rerun on 6 October. Counts reconcile automatically with the raw reports; no tests were skipped. The evidence supports the tested workflows; it does not establish production readiness.

Member	Student ID	SEF module
Jathurshy.S	IT24103355	Incident Management and Coordinator/Planner Agent
M.Thanusikan	IT24103609	Resource and Inventory Management and Situation Analysis Agent
Denesh N	IT24102826	Dispatch and Task Assignment and Resource Allocation Agent
Deneshkar. P	IT24103828	Volunteer/Responder Management and Validation/Safety Agent

Executed area	Passed	Failed	Skipped
Backend including PostgreSQL and migrations	95	0	0
React components and services	47	0	0
Flutter units and widgets	91	0	0
Python deterministic agents	20	0	0
Live HTTP workflow and security	17	0	0
k6 load scenario	1	0	0
Migration CLI check	1	0	0

Assessment brief: Software Testing and Quality Evaluation of the SE3090 Integrated System. Deadline: 5 October 2026. The source filename is SE3110 Assignment.pdf, while its internal module heading is SE3090; the group should confirm the CourseWeb location.

Test plan and environment

Objectives. Verify important implemented behaviour, demonstrate cross-component execution, measure a bounded performance workload, check authorization and safety controls, and preserve traceable defect/retest evidence. The same ERCP source is used throughout.

Scope. Incident reporting and planning; resources, inventory and shelters; dispatch and approval; volunteers and shifts; authentication; React UI states; Flutter models/widgets; deterministic agent contracts; PostgreSQL uniqueness, foreign keys, cascade deletion and rollback.

Area and tools	Reason for selection
xUnit, WebApplicationFactory, EF Core and PostgreSQL	Existing .NET framework; direct business/controller tests plus real HTTP authorization and actual relational constraints.
Vitest and React Testing Library	Existing setup; component rendering, interactions, route permissions and mocked API failure states.
flutter_test	Existing unit/widget fixtures; login validation, state, models and responder screens.
pytest and Requests	Actual Python agents and a live HTTP workflow against the real local API and PostgreSQL.
k6 1.3.0	Repeatable virtual-user workload with predeclared latency, failure and content thresholds.

Environment. Windows 11; .NET SDK 10.0.401 (net8.0 target); Node 22.15.0; Python 3.13.2; pytest 8.3.3; Requests 2.34.2; PostgreSQL 17.10 on 127.0.0.1:55432; API on 127.0.0.1:5009. Frontend build reported Vite 8.2.1. Flutter 3.44.9 uses its resolved lockfile; CI is pinned to that tested SDK. QMS mode requires PostgreSQL and applies EF migrations before accepting requests. Evidence records the actual tool outputs.

Isolation. A fresh ercp_qms_github database supports this full rerun. QMS startup rejects legacy tables without migration history and disables in-memory fallback. Earlier test databases are preserved. Migration tests create and remove only their own isolated databases. Most original backend tests still use EF InMemory. SMTP and Gemini credentials are disabled for live tests. No external messages or live model calls are required.

Schedule. Original analysis and testing were performed on 3 October. BUG-03 reproduction, repair, migration/backend/live workflow retesting followed by a complete reproducibility rerun and report revision were completed on 5 October. Integration with current main, publication and a full rerun were completed on 6 October. The brief states a 5 October deadline.

Risk based test design and responsibilities

Risk	Normal and failure or boundary coverage	Reviewer
Incorrect incident data or unsafe plans	Meaningful incident creation; invalid title/address/email and negative affected count; generated plan output and approval flag.	Jathurshy.S
Invalid resource or database state	Resource render/empty/error/status interaction; urgency levels; full shelter; exact capacity and stock limits; database unique/FK/cascade/rollback.	M.Thanusikan
Unauthorized dispatch or approval	Citizen/Volunteer mutation denial; invalid JWT; no approval before safety validation; immutable approved tasks; allocation shortages.	Denesh N
Unavailable or conflicting responders	Volunteer HTTP lifecycle; shift windows; adjacent shifts; unavailable volunteer rejection; malformed safety payload; mobile form/state tests.	Deneshkar. P

Entry criteria. Dependencies installed; isolated API and database reachable; original suites executed; valid fixtures available. **Exit criteria for the evaluated scope.** Record all outcomes, fix critical demonstrated authorization defects, verify retests, and identify remaining failures explicitly. BUG-03 now passes clean schema, rollback/data-preservation and upgrade checks. Adoption of an existing database created outside migration history remains a separate task.

Traceability. Test_Cases.html and test-cases.json contain 272 separately identified records: feature, preconditions, reproduction steps, expected assertions, actual result, status, source and evidence path. Parameterized variants are separate records. Expand the source excerpt to inspect exact fixture input and assertions.

Baseline and change. Before changes, backend 80/80 and React 39/39 passed. Seven new security regression cases produced four failures and three passes before the fix. The refreshed backend result is 95/95, including four new migration regressions. React 47/47 and Flutter 91/91 were rerun on 6 October. Existing superficial component tests remain visible and are not presented as strong behavioural coverage; the new resource and protected-route cases exercise actual components.

Ownership. The named allocation follows the user-supplied SEF ownership. Execution here was AI-assisted. Each member must review and run their contribution before claiming personal implementation or execution during the individual viva.

Integrated workflow and agent evaluation

E2E01 is a live API integration workflow. Requests drives the actual ASP.NET Core API backed by PostgreSQL. It is not a browser or physical-device end-to-end test.

Step	Observed check
1 Register citizen and report incident	201 responses; meaningful incident persisted using valid field inputs.
2 Generate response plan	Planner endpoint returned a plan for the created incident. Live LLM credentials were disabled.
3 Create responder and dispatch	Volunteer and proposed dispatch created; premature approval returned 400.
4 Assign task and validate	Task created; validation returned isValid=true and recommendation APPROVE.
5 Coordinator approves	Status update succeeded and subsequent GET returned Approved with the expected volunteer task.
6 Verify lock and persistence	Further task creation returned 409. Independent PostgreSQL query confirmed Approved.

Recorded fixture identifiers: incident 4, dispatch 1, plan 3. See evidence/final/workflow.json for the returned validation checks, saved dispatch and independent database verification. All 17 live HTTP cases passed again on 6 October against the migrated schema.

Agent evaluation. Twenty pytest cases import the actual four Python agents. Coverage includes rejected unapproved tools, coordinator structured output and mandatory human approval, allocation shortages, urgency categories, full shelters, exact capacity/stock boundaries, unavailable volunteers, malformed payload rejection and adjacent shift windows.

Injection scope. A malicious instruction string in an allocation title does not remove deterministic human approval. This only verifies that deterministic path. It is not a prompt-injection evaluation of live Gemini. Backend planner execution may use its implemented fallback; the report does not infer a live LLM response from a successful plan endpoint.

Environment recovery. The first live workflow was interrupted by Windows data-protection/event-log permissions. An authorized local rerun completed successfully. The initial failure is preserved as environment evidence and excluded from the application defect count.

Performance and security results

Performance design. The new 6 October 40-second k6 run ramps to 5 virtual users, reaches 10, then ramps down. Each iteration requests authenticated incident, dispatch and volunteer lists with a 0.2-second pause. Targets were p95 below 1000 ms, HTTP error rate below 1%, and all status/content checks passing.

Metric	Observed result
HTTP requests	3202
Throughput	79.70 requests/second
Response time p95	3.24968 ms
Maximum response time	70.90780 ms
HTTP failures	0.00%
Status and JSON checks	6402 passed; 0 failed
Thresholds	All three passed

These measurements describe a small local dataset and short authenticated read workload. They exclude mobile/browser rendering, live LLM latency, sustained stress, multi-machine deployment and production traffic. They must not be extrapolated into a production capacity claim.

Security execution. xUnit HTTP tests and pytest live HTTP tests verified public-role escalation rejection, low-privilege dispatch restrictions, malformed tokens, anonymous protected endpoints and a SQL-injection-shaped login input. The latter demonstrates rejection for the submitted input, not complete SQL-injection resistance. Coordinator success is verified through the integrated workflow.

Before and after. SEC01 initially allowed Admin and Coordinator registration (201); both now return 400. SEC02 initially let Citizen and Volunteer create dispatches (201); both now return 403. Live SEC05 additionally verifies forbidden task/status/allocation actions. Evidence is preserved in security-before.trx, backend.trx and live-api.xml.

Security limits. No ZAP scan was executed. The tested application intentionally exposes seeded demo credentials and has development-oriented configuration. These remain deployment risks. Successful authorization tests are not a complete security assessment.

Defects and quality assessment

Defect	Severity and status	Result
BUG-01 Public privileged registration	Critical / P1 Fixed and retested	DTO and public controller now reject privileged roles. Two role variants failed before and passed after.
BUG-02 Dispatch mutation permissions	High / P1 Fixed and retested	Coordinator/Admin restriction added to six mutation actions. Positive coordinator and negative low-privilege checks pass.
BUG-03 Overlapping migrations	High / P2 Fixed and retested	Duplicates and rollback ownership corrected; missing schema supplied by a new migration. Four migration regressions pass; the original failed database now upgrades.

Migration handling. The repaired history creates each table once. Rollbacks preserve earlier incident tables and resource data, and shelter foreign keys match the model. A new migration supplies Users, ValidationAudits and six missing planner fields. All eight migrations apply; the model snapshot has no pending changes. Original failures are retained alongside successful retests. Shared databases were not changed; startup-created databases require separate schema/history adoption.

Coverage. The final Coverlet report records 7567 of 9711 instrumented lines (77.92%) and 454 of 1434 branches (31.65%). Migration tests execute substantial generated/model code, so the increased line rate does not establish comparable business-logic coverage. Branch coverage remains limited. Frontend and Flutter coverage percentages were not collected; no percentages are inferred from test counts.

Static checks. Flutter analyze reports no issues. React lint reports zero errors and 4 warnings. The React production build succeeds with a bundle-size warning above 500 kB. Git diff whitespace checking passes. The team upgraded MailKit to 4.18.1 in commit cd96717. The current build has no earlier MailKit NU1902 warning; a comprehensive dependency audit is not claimed.

Remaining gaps. Physical devices, browser-driven complete workflows, live LLM failures, extended load/stress, accessibility scans, broad penetration testing and adoption of existing startup-created databases are outside the executed evidence. Existing mock-based tests must be interpreted within their isolation boundaries.

Defect_Report.html records reproduction steps, expected/actual behaviour, root cause, fix status, evidence and retest outcome for each application defect. Source changes are supplied with the application and evidence/final/security-fixes.patch and evidence/migration-fix/migration-fixes.patch.

Contribution evidence and viva preparation

Repository: github.com/Deneshkar/emergency-resource-coordination-platform

Published source: 9226a56, based on main 054fce7. PR #47 preserves newer team changes, including the duplicate-table repair and MailKit update. Contribution_Record.md links verified GitHub test contributions from all four members; GitHub_Publication.md records publication and CI status.

Member	Verified existing commit and suggested demonstration
Jathurshy.S - IT24103355	50f9414: Incident controller tests and planner implementation. Demonstrate invalid input and approval requirements.
M.Thanusikan - IT24103609	25de8a9: Resource, shelter and inventory edge-case tests. Demonstrate a PostgreSQL constraint and migration regression.
Denesh N - IT24102826	48a9c3b: Dispatch allocation/controller and React tests. Explain authorization, allocation shortages and load thresholds.
Deneshkar. P - IT24103828	3c059a6: Volunteer/shift HTTP tests. Demonstrate safety rejection and a Flutter widget case.

Individual evidence status. The group confirmed on 6 October that personal runs of the new QMS tests by Jathurshy.S, M.Thanusikan and Denesh N are not yet confirmed. Existing test authorship is verified separately. Current group execution and publication are AI-assisted; personal review and viva demonstrations remain each member's responsibility.

Each member should run and explain the fixture, assertion, tool setup and result; modify an input and predict the outcome; locate one defect and its retest evidence. The marking scheme assigns 40 group marks and 60 individual marks, so reading the report alone is insufficient preparation.

AI declaration. OpenAI Codex assisted repository analysis, test ideas and implementation, security fixes, execution, evidence reconciliation and report generation on 3 October, with migration repair on 5 October and integration, publication and full retesting on 6 October 2026. Generated tests were checked against the actual source and executed. Deterministic AI paths were used with live Gemini disabled. Members must review and understand all submitted work and record their genuine contribution.

The brief requires the module CLEAR framework but does not define its fields. Use this declaration as source content for the lecturer-provided CLEAR template; no alternate definition has been assumed. Contribution_and_Viva.md provides the fuller declaration and assigned rehearsal tasks.

Conclusion and evidence index

The tested incident-to-approved-dispatch workflow completes on the actual API and PostgreSQL, and selected resource, responder, client and agent behaviours pass their automated checks. The assessment corrected two authorization defects and the migration defect. The repaired schema passes clean installation, rollback/data-preservation and upgrade checks, and the workflow passes on that schema. The new local load test met its thresholds. The remaining security, coverage and deployment-adoption limits prevent a production-readiness conclusion.

Submission item	Purpose
Software_Testing_Report.pdf	Test plan, scope, environment, results, defects and conclusion.
Test_Cases.html and test-cases.json	272 completed, evidence-linked records with expected and actual results.
Defect_Report.html and defects.json	Three defects with severity, reproduction, evidence and retest status.
Contribution_Record.md, Contribution_and_Viva.md and GitHub_Publication.md	Verified commit links, publication/CI status, pending personal execution, viva guide and AI declaration.
Reproduction_Guide.md and scripts/run-tests.ps1	Dependencies, isolated environment setup and rerun commands.
evidence/baseline and evidence/security-before	Original suites and pre-fix security failures.
evidence/final/backend, react.json, flutter.jsonl, agents.xml, live-api.xml	Raw automated execution and backend coverage.
evidence/final/performance.json, workflow.json and migrations.log	Load metrics, migrated workflow evidence and passing CLI upgrade. Original failure and four regression retests are in evidence/migration-fix.
Application source and testing scripts	Reproducible implementation, configuration examples and test code.

Submission checks. Review member details; supply a group number if required; commit genuine member contributions and verify the repository link; transfer the AI declaration into the official CLEAR format; rehearse the individual demonstrations; upload through the correct CourseWeb entry. The QMS source is published in PR #47; CourseWeb upload has not been performed.

Source basis. The supplied four-page SE3110 Assignment.pdf, the ERCP repository at published source commit 9226a56 on top of main 054fce7, and the tool-generated evidence listed above. Historical report pass counts were not reused as fresh execution results.